

MotionForge 3D — Movement Analysis Report

Decision-support analytics only. This report is not a medical diagnosis.

Organization	MotionForge Public Dataset Lab
Subject	Wikimedia squat demonstration sample
Session	Public Dataset Weighted Squat Analysis
Movement template	squat
Analysis mode	camera_plane_3d
Generated	2026-07-19T06:54:05.328538+00:00

Data quality

Decoded Frames	213
Analyzed Frames	27
Valid Pose Frames	27
Valid Frame Percentage	100.0
Average Landmark Confidence	0.5372
Sample Fps	4.0
Source Fps	30.0
Duration S	7.1

Movement summary

Explainable score	80.08
Repetitions	2
Symmetry	88.19348516406464

Joint	Min	Max	ROM	Mean
left_elbow	1.0	47.9	46.9	18.2
right_elbow	7.8	53.9	46.1	36.2
left_shoulder	3.2	33.5	30.3	20.0
right_shoulder	7.9	26.2	18.3	18.0
left_hip	68.3	179.7	111.5	155.9
right_hip	58.4	174.1	115.7	135.1
left_knee	43.4	179.1	135.7	153.0
right_knee	58.9	178.5	119.7	138.2

Evidence



Limitations

Monocular measurements are camera-relative and not metric. Camera angle, lighting, loose clothing, occlusion, and overlapping people can reduce accuracy. Metric distances require calibration. Review generated findings before operational use.